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SEM: 5 SECTION: E

SOFTWARE ENGINEERING LAB 03

ARCHITECTURE SELECTION:

I chose Microservices Architecture for the Centralized Audit Trail Compliance Engine.

Reason 1: Different Load Patterns

The ingestion service keeps receiving audit logs continuously, while the query service needs to search across 10 million records in under a second. These two workloads are very different, so keeping them as separate services lets us scale each one on its own without either slowing the other down.

Reason 2: Independent Updates to Compliance Rules

Compliance officers need to add or change PII-masking rules often as new fields appear, but the core ingestion and hashing logic should stay stable. With microservices, the Masking Rules Config service can be updated on its own, without redeploying the whole system.

SECURITY ADVANTAGE:

Each service can have its own access control and its own encryption keys. So if one service, like the query service, is ever compromised, the attacker cannot directly reach the ingestion or hashing services. This limits the damage and helps meet GDPR's data protection requirements.

PERFORMANCE ANALYSIS:

Since the query service is separate, we can give it its own read replicas and caching to hit the sub-1-second search requirement. This way, heavy search traffic from auditors never competes with the constant stream of new logs being written by the ingestion service.

COMPONENT DIAGRAM:

